

Vertex Example

The diagram shows the sum of three Feynman diagrams equal to zero, representing a Ward identity. The first diagram shows a fermion line with a vertex correction: a wavy line (photon) connects a fermion line (with a triangle loop) to a vertex (with a W^- loop). The second diagram is the same as the first, but the vertex correction is on the other side of the fermion line. The third diagram shows a fermion line with a vertex correction: a wavy line (photon) connects a fermion line (with a triangle loop) to a vertex (with a W^- loop). The sum of these three diagrams is equal to zero.

$$\text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} = 0$$